Roll No._____ Name_____

DEPARTMENT OF MATHEMATICS, T. I. E. T., PATIALA

M.Sc. (Maths. and Computing) Session: 2024-25 (Even) Mid Semester Examination, 2025

Time: 2 Hrs. Numerical Analysis (PMA-208) March 12, 2025 Max. Marks: 30

Note: Answer all the questions. Course Coordinator: Dr. Arvind Kumar Lal

1.(a)	Consider the function $f(x) = \sqrt{x+1} - \sqrt{x}$ and $x = 12345$. Using six significant digit chopping arithmetic, show that this is a well- conditioned but not stable. Find an equivalent form of $f(x)$ so that it can be made stable way of evaluation. (3.5) [CO1, L2]
(b)	Determine $p,\ q\ and\ r$ so that the order of the iterative method $$x_{n+1} = p x_n + \frac{q\,\alpha}{x_n^2} + \frac{r\,\alpha^2}{x_n^5}$$ for $\alpha^{1/3}$ becomes as high as possible. For this choice of $p,\ q\ and\ r$, indicate how the error in x_{n+1} depends on the error x_n. (4.5) [CO2, L1]
2.(a)	Find the multiplicity of a root $\alpha = 0$ of the equation $\sin x^3 = 0$. Working with four decimal digits rounding arithmetic, perform only two iterations of modified Newton's method using $x_0 = 0.5$. (3.0) [CO2, L3]
(b)	Working with four decimal digit rounding arithmetic, perform only two iterations of Newton – Raphson's method to solve the system of nonlinear equations: $x^2 + xy + y^2 = 7;\ x^3 + y^3 = 9$ using initial approximations $x_0 = 1.5,\ y_0 = 0.5$. (4.0) [CO2, L3]
3.(a)	The equation $e^x - 4x^2 = 0$ has a root in $[4, 5]$. Show that we cannot find this root using fixed – point iteration method with iteration function $x = \frac{1}{2} e^{x/2}$. Can you find an iteration function, which will correctly locate the root? If yes, estimate the number of iterations required to achieve accuracy 0.005, starting with $x_0 = 4$. (4.0) [CO3, L3]
(b)	Obtain a formula to compute total number of arithmetical operations required by Gauss elimination method to solve the linear system of equation $Ax = b$, A is square matrix of order n. (3.0) [CO3, L3]
4.(a)	Consider the following linear system of equations: $2x_1 - x_2 + x_3 = -1;\ 2x_1 + 2x_2 + 2x_3 = 4;\ -x_1 - x_2 + 2x_3 = -5$ Compute the spectral radius of both Jacobi's and Gauss – Seidel iteration matrices and analyze convergence of both the methods. (4.5) [CO3, L2]
(b)	The linear system $1.003\,x + 58.09\,y = 68.12;\ 5.550\,x + 321.8\,y = 377.3$ has actual solution $X = [10,\ 1]^T$ and approximate solution $\bar{X} = [-10.0,\ 1]^T$. Compute (i) $\|X - \bar{X}\|_\infty$ (ii) $K_\infty(A)\ \frac{\|b - A\bar{X}\|_{I\infty}}{\|A\|_\infty}$ (iii) Is there any relationship between (i) and (ii)? Justify your answer. (3.5) [CO3, L2]

THE END